



**Health & Safety**

# Standard Hazardous chemicals



Reference

| Company | Document Type | Department | N°   | Version | Langue |
|---------|---------------|------------|------|---------|--------|
| BYTP    | REF           | H&S        | 2222 | B       | EN     |

| Version | Date       | Write by   | Checked by                 | Approved | Reason for modification                                                     |
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| A       | 01-11-2023 | A.BERNARD  | I.LELIEVRE                 | L.KNOLL  | First issue                                                                 |
| B       | 21-05-2024 | A.JANSSENS | I.LELIEVRE<br>O.BRZEZINSKI | L.KNOLL  | Removal of 1. H&S basics<br>Modification of section 1<br>Safety Data sheets |
| C       | 22-10-2024 | F.MARTIN   | I.LELIEVRE<br>O.BRZEZINSKI | L.KNOLL  | Modification of Section 11<br>Silica following BYCN<br>requirements update  |

The standards for the management and use of hazardous chemicals presented in this document serve as a practical guide to define the means to plan and deploy.

The elements of this document is complemented by the elements presented in the following MOSE+:

- 5711 – Implementation of sprayed concrete by dry process
- 5737 – Installation of carbon fiber plates and fabrics
- 5753 – Application of wet shotcrete
- 5882 – Planing

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# 1. Safety data sheets

The SDS provides all the information on the product: components, dangers and preventive measures. Always comply with the instructions and recommendations of the Safety Data Sheets (SDS), including wearing specific personal protective equipment if required.

The data sheet is the instructions for use of the product.



Explosive



Flammable



Oxidizer



Gas under pressure



Corrosive



Poisonous/deadly



Harmful / Health impairment



Toxic / Harmful to health



Dangerous for the environment

## 2. Preventive measures

### Substitution

- From the consultation phase replace, when possible, with non-hazardous or less hazardous products

### Fire safety

- Keep a fire extinguisher in the vicinity of the workstation
- Comply with the safety instructions



### Collective protection and organization

- Create general ventilation and vacuum at source when required
- Limit the number of people in the area

In the presence of flammable or explosive products, it is forbidden to smoke or to generate a source of ignition

### Individual protection

- Wear appropriate personal protection: boots, gloves, goggles, cartridge mask, coveralls



## 3. Carcinogens mutagens reprotoxic (CMR) agents

Some chemical agents have, in the medium or long term, carcinogenic, mutagenic or reprotoxic effects. They are called CMR agents. It is essential to identify them, ie to make an inventory of the products used and the work situations that may give rise to exposure. When a CMR agent is spotted in the workplace, his removal or substitution is necessary, whenever technically possible.

The use of category 1 CMR products (excluding fuels) is prohibited.  
All products are subject to approval by the Occupational Health Service.

## 4. Product storage

- Comply with container identification rules
- Never mix the products if it is not intended for this purpose
- Respect the compatibility of products for their storage (see table below) and SDS, including in laboratories
- The installations are adapted to the physico-chemical characteristics of the products (ventilation, sealed cabinets, weather protection, extinction means, etc.)



No storage of chemicals in vehicles cabins

|  | + |   |   |   |   |   |   |   | + |
|--|---|---|---|---|---|---|---|---|---|
|  |   | + |   |   |   |   |   |   | + |
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|  |   |   |   |   |   |   | + | + | + |
|  |   |   |   |   |   |   |   | + | + |

+: Authorized    O : under conditions (check SDS)    In red: Prohibited



Retention pallet tray



No jerrycan storage in direct sunlight

## 5. Product referencing

The Safety Data Sheets (SDS) of all the chemicals available on site (including those of the subcontractors) are necessarily:

- available on site
- transmitted to the nursing staff if present

Users are trained in their use, based on the SDS.

The inventory of hazardous products is forwarded annually to the medical department.





## 6. Use of chemicals

**The use of any chemical product requires a training specific to the workstation.**

Some products require specific PPE to be worn (refer to the SDS). If in doubt, ask the Occupational Health Service or the Health and Safety Department.

Transfer = Identification of containers

The use of category 1 CMR products (excluding fuels) is prohibited.  
All products are subject to approval by the Occupational Health Service.



## 7. Concrete handling

Handling concrete requires wearing specific PPE (see PPE guide), in addition to the mandatory PPE:

- Waterproof gloves
- Safety rubber boots



## 8. Disposal of waste

**Using chemicals consistently means:**

- Plan and anticipate their elimination
- Storage in bins provided for this purpose
- Identify bins for disposal



## 9. Soil treatment

Soil treatment is an operation that consists of mixing natural soil with lime and/or a hydraulic binder. It allows, by improving its characteristics, to use a ground which was unsuitable in the natural state.

### Exposure risks

Quicklime is dangerous due to its alkaline nature (like fresh cement): irritating to the mucous membranes and the respiratory tract, it can cause chemical burns. Indeed, quicklime is a caustic product.



In the presence of water, it hydrates with a release of heat that can reach 80°C and thus be the cause of thermal burns.

In addition, the very fine grain size of quicklime allows it to disperse easily in the air and facilitates contact with site personnel and people outside the site depending on the conditions.

### Main prevention measures

- SDS on site MANDATORY
- Have suitable equipment: equip and maintain soil treatment machinery, choose and use treatment equipment that limits the dispersion of lime
- Organize the soil treatment workshop
- Prepare lime loading and storage stations
- Use appropriate PPE: overalls, cotton-lined gloves with cuffs, dust respiratory protection
- Adopt strict hygiene measures: remove soiled clothes and take a shower at the end of the shift

### In case of exposure

Consult a doctor immediately

#### If inhaled

Move source of dust or affected person to fresh air.

#### In case of skin contact

Carefully and delicately brush the contaminated parts of the body in order to eliminate all traces of the product. Wash affected area immediately with plenty of water. Remove contaminated clothing.

#### In case of eye contact

Flush eyes with plenty of water.

#### If swallowed

Rinse mouth with water, then drink plenty of water.  
Do NOT induce vomiting.



Portable emergency shower  
(annual check + supplier  
recommendations)

## 10. Work in waste collection, sewers and WWTP

### Risk assessment

The risk assessment leads to an inventory of the materials, products or work processes likely to generate risks for the health of employees:

- Technical characteristics of the works concerned
- Types of effluents and, in particular, nature of spills that may occur
- Nature and results of the atmosphere checks carried out, in particular with regard to the oxygen level, the presence of gaseous pollutants, explosiveness, H<sub>2</sub>S, CO, CO<sub>2</sub>
- Nature and results of effluent controls

It is then necessary to identify the conditions under which employees could be exposed, then to assess the levels of exposure.

Rehabilitation work on an occupied site



### Main prevention measures

- Define the method statement for each intervention
- Provide showers in sanitary facilities
- Reinforced medical surveillance
- Vaccinations and reminders to be included in the medical file (Occupational Health Service)



Showers on site

Depending on the conditions of the site, work in landfills, sewers and WWTPs can be classified as work in confined spaces. If necessary, it is essential to integrate the associated organizational and prevention measures into the operating methods.

Refer to document **BYTP-REF-H&S-2223 (Standard Confined space)** .

# 11. Silica

## The risk of exposure to silica

The risk assessment leads to an inventory of the materials, products or work processes likely to emit crystalline silica dust. It is then necessary to identify the conditions under which employees could be exposed, then to assess the levels of exposure.

As a reminder, the use of cement is the main source of silica on our worksites.

## Main prevention measures

- Use closed systems that are as airtight as possible (eg. shredders and carefully covered conveyor belts in quarries)
  - Favor work methods that generate little or no dust, such as wet work
  - During earthworks, ensure analyzes are performed by laboratory teams
  - Work involving exposure to silica dust is carried out in segregated areas with restricted access (sanding cabin)
  - Equip workstations with a dust source capture device (device integrated into the process or tooling with or M extraction unit as per EN 60335-2-69 or an equivalent) when operations cannot be carried out in a closed system
  - Regularly check the operation of the ventilation devices
  - When necessary, provide personnel with suitable goggles and type 5 disposable hooded coverall
  - Unless specific measurements taken in the area demonstrate that exposure is below the thresholds defined by local regulations (when no local regulations are available, refer to the most applicable international regulations), it is mandatory for anyone working within 2 meters of the task (in an open environment) or within a 100 square meter area (in a closed environment) to wear:
    - a half mask or FFP3 mask if the daily duration is less than 1 hour
    - a powered air-purifying respirator if the daily duration is more than 1 hour
  - Regularly check the dustiness of the working atmosphere and at each major change in operating methods
  - Perform regular cleaning of workplaces (vacuum equipped with a very high efficiency filter or wet cloths)
  - Store and wash work clothes separately from other clothes
  - Ensure the proper awareness and training of the personnel
  - Medical surveillance of exposed employees and post-activity follow-up
  - Pickling, sand removal and jet etching (see regulations)
- = specific protective measures





## 12. Silica – Threshold Limit Values

Silica is among the carcinogens for which the regulations in force must be respected. It is appropriate to identify the occupational exposure limit value (OEL) – Average concentrations over 8 hours – of:

- Inhalable dusts
- Alveolar dust
- Silica (quartz)
- Silica (cristobalite and tridymite)

In the absence of limit values in local regulations, refer to the most applicable international regulations.

As an example, French regulation is mentioned below:

|                                     | OEL (Occupational Exposure Limit Values)<br>Average concentrations over 8 hours |
|-------------------------------------|---------------------------------------------------------------------------------|
| Inhalable dust                      | 4 mg/m <sup>3</sup> of air                                                      |
| Alveolar dust                       | 0.9 mg/m <sup>3</sup> of air                                                    |
| Silica (quartz)                     | 0.1 mg/m <sup>3</sup> of air                                                    |
| Silica (cristobalite and tridymite) | 0.05 mg/m <sup>3</sup> of air                                                   |

# 13. Asbestos

## The risk of exposure to asbestos

All interventions on elements containing asbestos are:

- Subject to the provisions of local regulations
- To be carried out by a specialized company
- Before any intervention on asbestos, contact the Health and Safety Department

Working with a diagnosis is essential and it is necessary to check the adequacy between the diagnosis and the planned work.

Removal and containment work must be subcontracted to a certified company



In France, the asbestos diagnosis is provided by the Client. Please refer to the local regulation.

## Preventative measure

When the work is likely to release asbestos fibres, it is necessary to:

- Define specific asbestos operating procedures (Refer to the local regulation for any required transmission of information to the local authorities)
- Isolate work areas with asbestos materials from other work areas (dust) including locker rooms and showers
- Deploy dust measurements in the air
- Confirm the ability of staff to perform the planned operations and set up a proper medical follow-up
- Use specific PPE depending on the type of intervention (duration, dust, etc.)

## Trainings

|  | Type of activity | Minimum duration of prior training | Minimum duration of first refresher training (six months after the preliminary training) | Minimum duration of refresher training (three years after previous refresher training) |
|-------------------------------------------------------------------------------------|------------------|------------------------------------|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Technical support staff (work supervisor)                                           | Withdrawal       | 10 days                            | 2 days                                                                                   | 2 days                                                                                 |
|                                                                                     | Exposure         | 5 days                             | Not applicable                                                                           | 1 day                                                                                  |
| Site supervision staff (Masters, CE)                                                | Withdrawal       | 10 days                            | 2 days                                                                                   | 2 days                                                                                 |
|                                                                                     | Exposure         | 5 days                             | Not applicable                                                                           | 1 day                                                                                  |
| Operator (companions)                                                               | Withdrawal       | 5 days                             | 2 days                                                                                   | 2 days                                                                                 |
|                                                                                     | Exposure         | 2 days                             | Not applicable                                                                           | 1 day                                                                                  |

## 14. Lead

### The risk of exposure to lead

All interventions on elements containing lead are:

- Subject to the provisions of local regulations
- To be carried out by a specialized company
- Before any intervention on lead, contact the Occupational Health and Safety Prevention Department.

Working with a diagnosis is essential and it is necessary to check the adequacy of the diagnosis/planned work

In France, the diagnosis is provided by the Client.



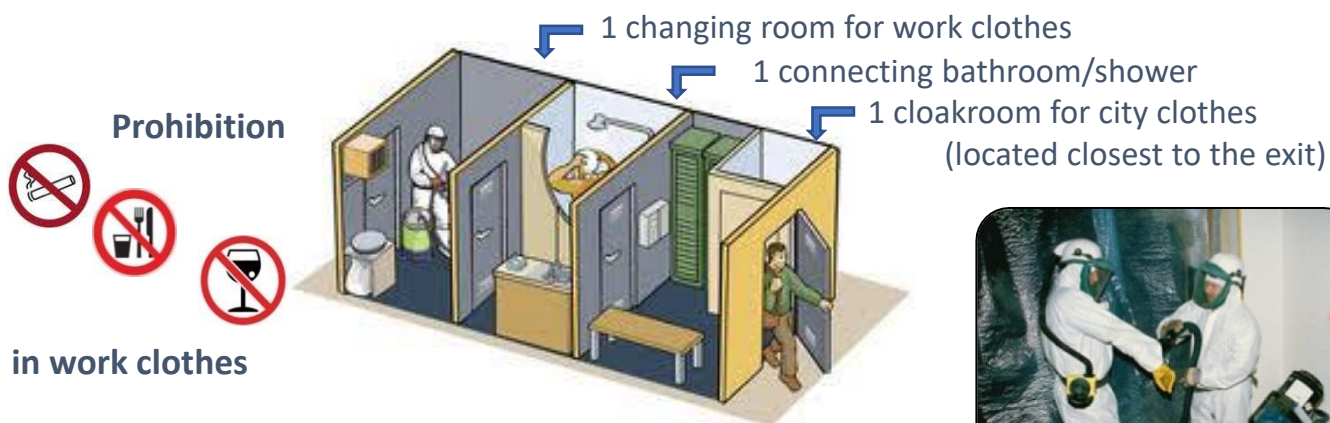
**Not formed = not touch**

Interventions with a naked flame (torch) are to be avoided



### Lead site" fittings

Workers exposed to lead have specific, isolated rooms:



- Aptitude and reinforced medical follow-up
- Specific PPE depending on the type of intervention
- Intervention by slice of 2h30 max.
- Avoid the emission of lead vapor by favoring "cold" work or with water supply
- In case of dust emission:
  - Watertight containment with access lock
  - Vacuuming of work clothes and waste bags before leaving the work area

